THE BIT DELIVERED TWICE: TWO KINDS AND NO THIRD

STEP 10 · THE BIT DELIVERED TWICE, BY ITS TWO KINDS, AND NO THIRD

KIND 1 · THE FORMAL CARGO

The record's value carried to the host's frame. It arrives free, the antecedent a theorem; it gives the Poincaré sentence in one step; and it is exactly the sentence.

`FormalCargo` · `cargo_arrives` · `formal_kind_delivers`
`formal_iff_sentence`

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KIND 2 · THE PHYSICAL WITNESS

A physical relaxation of every manifold of the host, faithful to the class, every step off the seat priced at the Landauer floor, bringing every manifold to rest on the seat. Its last field is the bit read in the register of heat, carried as a field, never proved.

`PhysicalWitness` · `witness_closes` · `physical_kind_delivers`
`physical_iff_sentence`

The two kinds are one bit
The physical witness is a formal cargo, and the cargo is exactly the sentence.
`the_two_kinds_are_one_bit`

Two kinds and no third
Every witness of the sentence, in any vocabulary, is a delivery of both kinds.
`two_kinds_and_no_third`
`every_route_is_both_kinds`

The one refuter
One manifold of the host that is not the round sphere, one fake sphere, refutes both kinds at once.
`the_one_refuter` · `refuter_denies_the_sentence`

THE 2002–03 OFFERING, TYPED · ONE INHABITANT OF THE FORMAL KIND, DOWNSTREAM OF THE BRIDGE

One inhabitant of the formal kind
A witnessed flow, faithful to the class with a functional strictly decreasing off the seat, delivers the formal cargo.
`WitnessedFlow` · `witnessed_flow_delivers_cargo`

Strictly inside the cargo
The cargo stands where the host's own classifying flow admits no witness: the offering is never the ground.
`cargo_without_witness`
`witness_downstream_strict_subtype`

It walks our way
Every witnessed flow delivers both kinds, and fails if the bit fails: our way in its own vocabulary.
`the_offering_typed` · `the_offering_walks_our_way`

READING theorem on no axiom at all · binding theorem on Lean's standard axioms · definition or structure · names shown by their last component